



GETTING STARTED · BACKDROPS

Open Scratch with a fun outdoor backdrop ready to go



STEPS TO FOLLOW

1. Go to scratch.mit.edu and click "Create"
2. Right-click the cat sprite → Delete
3. Click "Choose a Backdrop" (bottom-right)
4. Search "Blue Sky" or "Garden" → click to add
5. Sign in to auto-save your project!

TIP

Create a free Scratch account so your project saves automatically — look for "Sign in" at the top right.

DID YOU KNOW?

The Scratch stage is 480×360 pixels.

$x=0$, $y=0$ is the very centre. Positive y
= up, negative y = down.



SPRITES · COSTUMES

Add a character sprite then customise its look in the Costumes editor



STEPS TO FOLLOW

1. Click "Choose a Sprite" → pick a character
2. Click the **Costumes** tab (top-left)
3. Use the Fill tool  to recolour body parts
4. Use Brush/Shape tools to add details
5. Duplicate costume → tweak legs for walk animation
6. Click "Code" tab → set Size to 60

TIP

Duplicate a costume 2–4 times with small changes to make a walking animation. Add "next costume" inside the forever loop!

DID YOU KNOW?

Every Scratch sprite image is called a "costume". Switching costumes fast creates animation — the same trick used in cartoons and flipbooks.



EVENTS · MOTION · CONTROL

Left and right arrow keys move your character — and the screen edge stops them



SCRATCH BLOCKS

when  clicked

go to x: (-150) y: (-120)

forever

if <key [left arrow] pressed?> then

change x by (-5)

if <key [right arrow] pressed?> then

change x by (5)

if on edge, bounce

TIP

Set rotation style to "left-right" in

Sprite Info so "if on edge, bounce"

never flips the character upside-down!

DID YOU KNOW?

"if on edge, bounce" is a shortcut for:

check x position, if past the edge

reverse direction — it saves writing 4

extra blocks yourself.



LOOPS · EVENTS

Press Space to make the character rise then fall back to the ground



ADD THIS NEW SCRIPT TO YOUR CHARACTER SPRITE

when [space▼] key pressed

if <(y position) < (-115)> then

repeat (10)

change y by (12)

end

repeat (10)

change y by (-12)

end

end

TIP

The outer "if" stops double-jumping.

Try changing 12 to 8 or 16 to make the jump shorter or higher!

DID YOU KNOW?

This jump is linear — it rises and falls at the same speed. Step 5 upgrades it to feel natural, just like real games!



VARIABLES · PHYSICS

Upgrade the jump so gravity pulls the character back — just like a real platformer



REPLACE THE SPACE-KEY SCRIPT WITH THIS UPGRADE

```
when  clicked
  set [y velocity▼] to (0)
  forever
    if <key [space▼] pressed?> then
      if <(y position) < (-115)> then
        set [y velocity▼] to (12)
      end
    end
  end
  change [y velocity▼] by (-1)
  change y by (y velocity)
  if <(y position) < (-120)> then
    set y to (-120)
    set [y velocity▼] to (0)
  end
end
```

TIP

"change y velocity by (-1)" is gravity

— it runs every frame so the character accelerates downward naturally.

DID YOU KNOW?

Mario, Sonic, Celeste, Hollow Knight

— they all use this exact trick. You just coded real game physics!



MULTIPLE SPRITES · POSITIONING

Place an Apple sprite on the ground at a fixed position



SCRATCH BLOCKS — ON THE APPLE SPRITE

when  clicked

go to x: (0) y: (-120)

TIP

Always check which sprite is selected
(blue outline in the sprite panel)
before coding — each sprite has its
own scripts!

DID YOU KNOW?

In real game engines like Unity and Godot, every character, item, and effect is a separate object with its own code — just like each Scratch sprite.



OPERATORS · RANDOM NUMBERS

Make the apple appear at a different spot each time using "pick random"



SCRATCH BLOCKS — ON THE APPLE SPRITE

when  clicked

go to x: (pick random (-180) to (180)) y: (-120)

TIP

The y stays fixed at -120 (the ground level) — only the x is random so the apple always lands on the floor!

DID YOU KNOW?

Random number generation powers loot drops, enemy spawns, procedural maps, weather systems, and nearly every game mechanic that feels "alive".



SENSING · FOREVER LOOP

Make the apple react with a speech bubble the moment the player touches it



SCRATCH BLOCKS — ON THE APPLE SPRITE

when  clicked

go to x: (pick random (-180) to (180)) y: (-120)

forever

if <touching [your character]?> then

say [ Got it!] for (0.5) secs

TIP

"touching?" checks pixel-by-pixel

overlap 30 times per second inside

the forever loop. That's called polling!

DID YOU KNOW?

Collision detection is one of the most important algorithms in games. Unity calls them "colliders", Godot calls them "collision shapes" — same idea.



VARIABLES • STATE

Create a Score variable that increases by 1 each time an apple is collected



SCRATCH BLOCKS — ON THE APPLE SPRITE

when  clicked

set [Score] to (0)

go to x: (pick random (-180) to (180)) y: (-120)

forever

if <touching [your character]?> then

change [Score] by (1)

TIP

Tick the checkbox next to "Score" in the Variables palette to show/hide the score display on stage. Right-click → "Large readout" for big numbers!

DID YOU KNOW?

A variable is the most fundamental idea in all programming. Python, JavaScript, Swift, C++ — every language uses them to remember and update information.



GAME LOOP · RESPAWN PATTERN

After collecting, teleport the apple to a new random spot so the game never ends



SCRATCH BLOCKS — ON THE APPLE SPRITE

when  clicked

set [Score] to (0)

go to x: (pick random (-180) to (180)) y: (-120)

forever

if <touching [your character]?> then

change [Score] by (1)

go to x: (pick random (-180) to (180)) y: (-120)

TIP

Game complete! Try adding more

apples, a countdown timer, or

different point values to increase the challenge.



DID YOU KNOW?

"Game loop" is this pattern: check

input → update state → render →

repeat. Every game in history — from

Pong to Fortnite — runs on this same

cycle.